Weekly Progress: Validation, Resampling, and Decision-Level Explanation

June 3, 2026

Executive summary. This week I closed the loop from implementation validation to empirical variability and decision-level explanation. First, the Step1c SPO+ code was validated against PyEPO references on both shortest-path and KEP oracles. Second, subset-resampling boxplots summarize the distribution of 2stage and SPO+ behavior under fixed graph pools. Third, decision-level analysis on Step2b d8 suggests that 2stage can remain competitive because many large edge-level errors are not decision-critical, while SPO+ helps when errors fall on decision-changing edges.

1 Overview

This week's work follows the three-step plan discussed after the previous meeting. First, I used PyEPO as an external reference implementation to check whether my current codebase contains implementation bugs. Second, I ran subset-resampling experiments to summarize variability with box plots. Third, assuming the implementation is reliable, I started a decision-level analysis to understand why the two-stage model can sometimes remain close to decision-focused methods such as SPO+/FY in decision quality. This report analyzes SPO+ first; FY remains a planned extension.

The resulting workflow is:

Step	Goal	Current status
1	Implementation validation	Passed for LR; SPO+ algebra/gradient/trajectory validated
2	Subset-resampling variability	Box plots generated on fixed graph pools
3	Decision-level explanation	First-pass analysis completed on Step2b d8

2 Step 1: Implementation Validation with PyEPO Reference

2.1 Motivation

The goal of this step is to check whether my implementation contains subtle bugs. The logic is simple: if my code cannot reproduce PyEPO behavior on a standard benchmark or on the same KEP oracle, then there may be a bug in the loss, gradient, optimizer update, solver interface, or metric calculation.

Table 1: Mapping this week's work to the advisor's three requested directions.

Advisor direction	What was done this week	Status
Reproduce shortest-path / check bugs	PyEPO-vs-my shortest-path LR and SPO+ checks; KEP Phase 0/1/2 PyEPO-reference validation.	Mostly complete
Run resampling experiments and boxplots	Fixed graph pool, fixed label regime, subset-seed resampling at train size 50; heldout400 boxplots.	First layer complete
Small-graph decision comparison	Step2b d8 decision replay, edge criticality, case studies, near-tie analysis, and decision-critical MSE correlations.	First-pass complete

2.2 Shortest-Path Benchmark

I first used PyEPO's fixed-data shortest-path setup as an external benchmark. For the two-stage LR baseline, my implementation fully matches PyEPO across all fixed-data shortest-path settings:

Table 2: Shortest-path PyEPO-vs-my comparison.

Quantity	PyEPO LR vs my LR	PyEPO SPO+ vs my SPO+
Result files	24 / 24	24 / 24
Rows	240 / 240	240 / 240
Passing settings	24 / 24	19 / 24
True SPO global max diff	3.12×10^{-19}	3.61×10^{-4}
Unambiguous SPO global max diff	2.07×10^{-19}	3.61×10^{-4}
MSE global max diff	1.09×10^{-47}	2.70×10^{-3}
Epoch diff	0	0

For SPO+, the loss, gradient, and one-step update match PyEPO when the oracle solution is shared. The remaining full-CSV differences are explained by shortest-path tie-breaking: the same cost vector can have multiple optimal paths, and PyEPO's `optDataset` can store a different optimal solution vector across repeated solves even when the optimal objective is identical. Since SPO+ gradients depend on the solution vector and not only on the objective value, this can create training divergence without implying an algebraic bug.

2.3 KEP Reference Validation

I then moved the validation to the actual KEP oracle. This is more directly relevant to Step1c because KEP is a reward-maximization problem with instance-specific graph structure. The KEP validation uses the same `CachedHybridKepModel` / Gurobi oracle and compares PyEPO-style SPOPlus reference computations with my Step1c reward-max SPO+ implementation.

2.4 Conclusion from Step 1

The implementation check supports the conclusion that the Step1c SPO+ algebra, reward-max sign convention, gradient scaling, optimizer update, KEP oracle interface, and evaluation path are consistent with an independent PyEPO-based reference implementation. The remaining shortest-path SPO+ CSV differences are attributable to oracle tie-breaking among multiple shortest paths, not to an identified error in the Step1c SPO+ formula.

Table 3: KEP PyEPO-reference vs Step1c validation summary.

Validation phase	Key comparison	Result
Phase 0 preflight	reward-max / cost-min sign adapter	all max diffs = 0
Phase 1 small setting	SPO+ forward loss	max diff = 2.9998×10^{-6}
Phase 1 small setting	SPO+ gradient and one-step SGD update	max diff = 0
Phase 2 full trajectory	$\theta_1, \theta_2, \|\theta\|$	max diff = 0
Phase 2 full trajectory	train SPO+ loss	max diff = 3.2975×10^{-6}
Phase 2 full trajectory	validation SPO+ loss	max diff = 2.2324×10^{-6}
Phase 2 full trajectory	train / validation decision gap	max diff = 0

3 Step 2: Subset-Resampling Variability

3.1 Motivation

The second step is to turn a single-seed observation into distributional evidence. The current resampling experiment intentionally isolates one source of randomness: the training subset. The graph pool, label regime, validation set, heldout set, and training initialization are fixed. Only the subset seed used to sample 50 training graphs is varied.

Thus, this experiment should be interpreted as:

Train-subset resampling on a fixed graph pool and fixed label regime.

It does not yet test graph-generation seed, label-noise seed, or training initialization seed.

3.2 Experimental Setup

The first-pass resampling setting is:

Item	Setting
Train size	50
Randomness varied	subset seed only
Graph pool	fixed
Training seed	fixed
Evaluation	heldout400
Main methods	2stage selected by validation MSE; SPO+ selected by validation SPO+ loss
Primary metric	mean normalized decision gap

3.3 Generated Plots

The current plotting script summarizes the heldout400 distribution with four main outputs:

- overall gap boxplot;
- by-block gap boxplot;
- paired reduction boxplot;
- relative reduction by block.

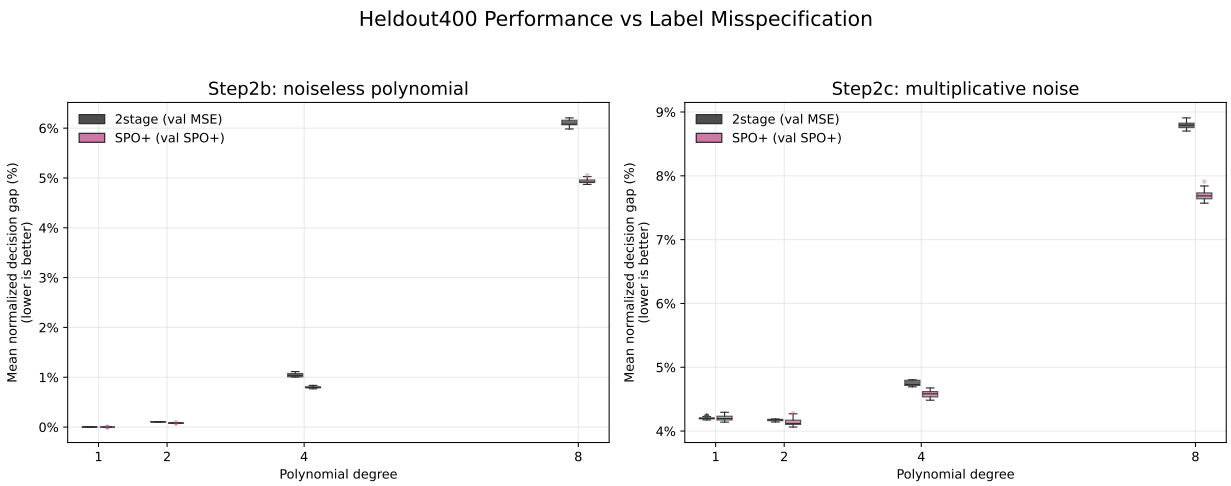

Figure 1: Heldout400 normalized decision gap by block under subset resampling.

How to read. Each box summarizes heldout400 normalized decision gaps across subset seeds for one regime/block and method. Lower boxes indicate better downstream decision quality; wider or higher boxes indicate stronger subset sensitivity.

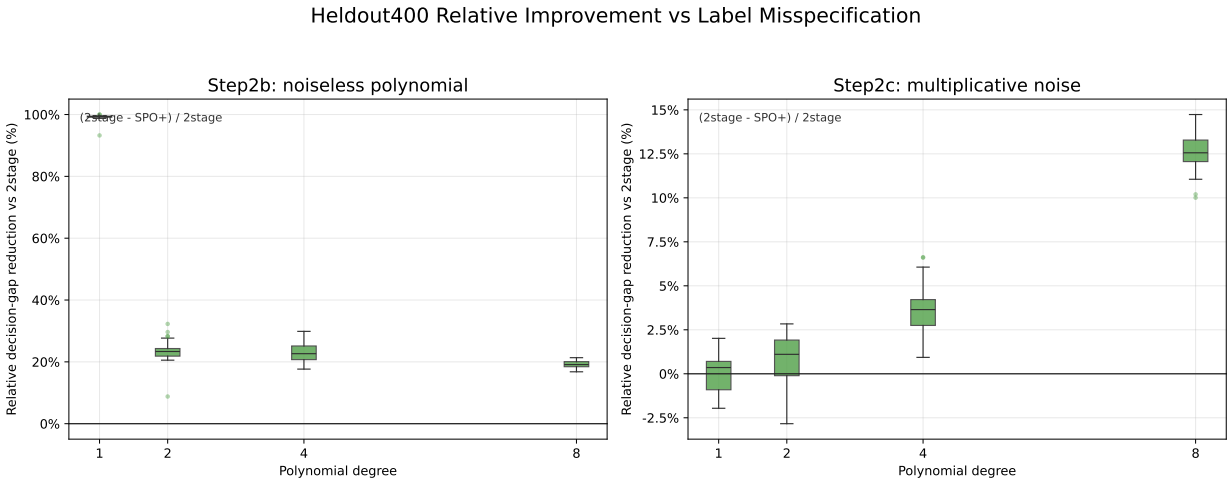

Figure 2: Heldout400 relative reduction versus 2stage by block.

How to read. Positive values mean the decision-focused method reduces normalized gap relative to the 2stage baseline on the same subset-seed protocol. Read larger positive boxes as stronger paired improvement, and values near zero as methods behaving similarly.

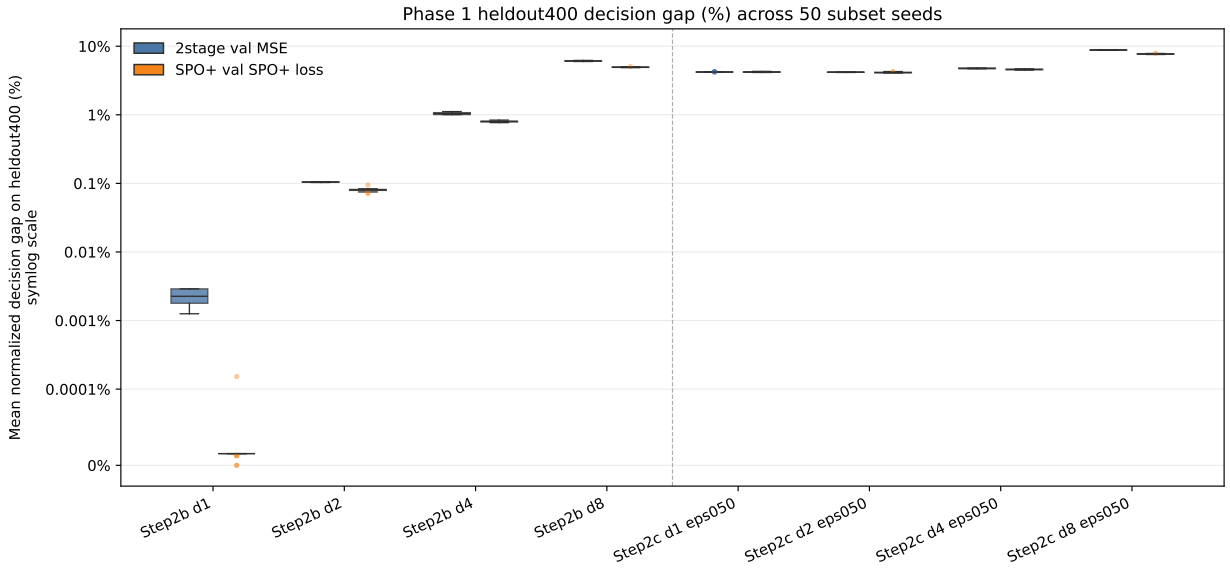

Figure 3: Overall heldout400 normalized decision gap distribution.

How to read. This plot aggregates the heldout400 gap distribution across the selected subset-resampling settings. Compare method medians and tails: lower medians indicate better typical quality, while shorter upper tails indicate fewer hard cases.

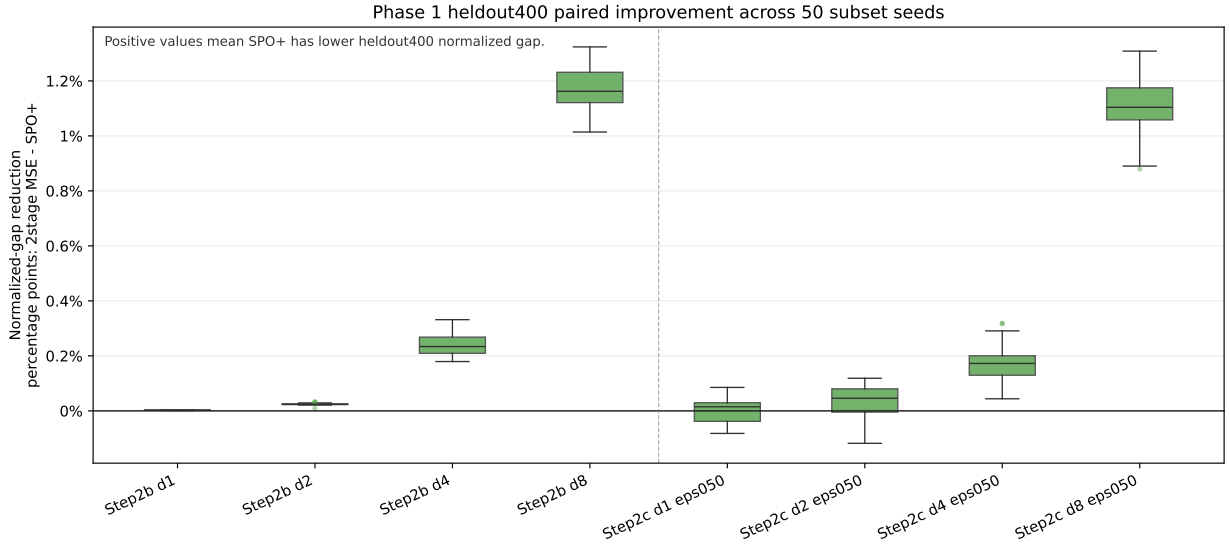

Figure 4: Paired heldout400 improvement: 2stage gap minus SPO+ gap.

How to read. Each value compares 2stage and SPO+ on the same subset seed. Positive values mean SPO+ has a lower normalized decision gap; the paired view reduces confounding from subset difficulty.

3.4 Interpretation

These plots answer whether the SPO+ improvement is stable under different sampled training subsets from the same fixed graph pool. The paired reduction plot is especially important because it compares 2stage and SPO+ on the same subset seed, reducing confounding from subset difficulty.

The correct conclusion is limited but useful:

> SPO+ improvement appears stable under train-subset resampling on the fixed Step2 graph pool, especially in high-degree misspecification regimes.

The current experiment should not be described as full robustness over all randomness. A stronger next-stage robustness test would vary graph-generation seed, label-noise seed, or training initialization seed.

4 Step 3: Decision-Level Explanation

4.1 Motivation

After validating the implementation and observing subset-resampling variability, I investigated why 2stage can remain competitive with SPO+. The hypothesis is that not all edge-level prediction errors are decision-critical. Some large errors occur on edges that never enter the selected KEP solution, while other selected solutions may be near-optimal alternatives.

4.2 Scope

The first-pass decision analysis focuses on:

$$\texttt{Step2b d8}, \qquad n_{\text{train}} = 50, \qquad \texttt{heldout400}.$$

It compares:

$$\text{2stage selected by validation MSE} \quad \text{vs.} \quad \text{SPO+ selected by validation SPO+ loss.}$$

Nine representative subset seeds were selected:

$$\{1, 25, 22\} \text{ large improvement}, \quad \{27, 21, 30\} \text{ weak/borderline}, \quad \{41, 16, 33\} \text{ easy low-gap}.$$

4.3 Data Products

The analysis produced:

Output	Rows / files
selected case seeds	9 rows
per-graph decision comparison	7,200 rows
edge-level criticality table	879,642 rows
graph-level criticality summary	7,200 rows
case-study index	9 rows
case-study edge tables	9 tables
margin / near-tie analysis	3,600 rows
observed candidate ranking	10,800 rows
decision-critical MSE correlations	12 rows

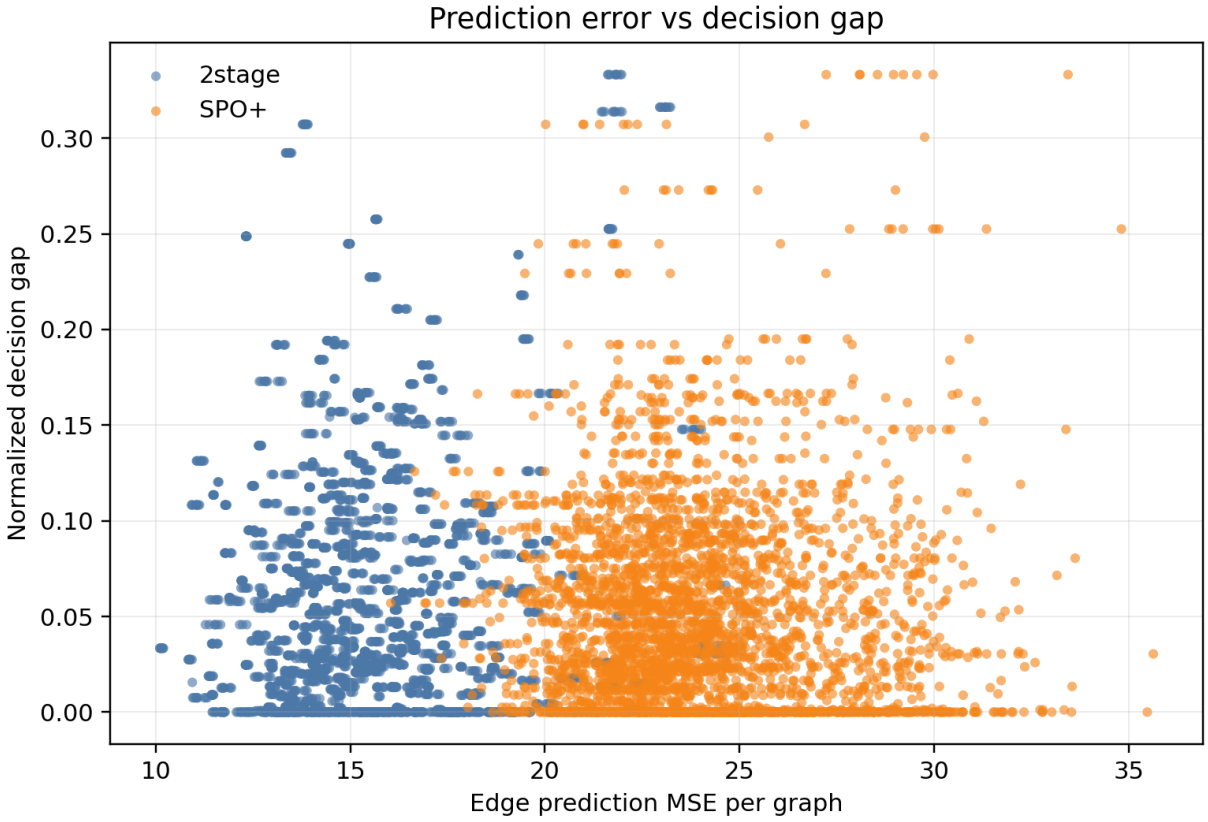

Figure 5: Prediction MSE versus normalized decision gap.

How to read. Each point is a method/graph evaluation from the selected Step2b d8 heldout400 cases. If raw prediction MSE fully explained decision quality, points would align strongly upward; the weak pattern shows that edge error alone is not enough to explain KEP regret.

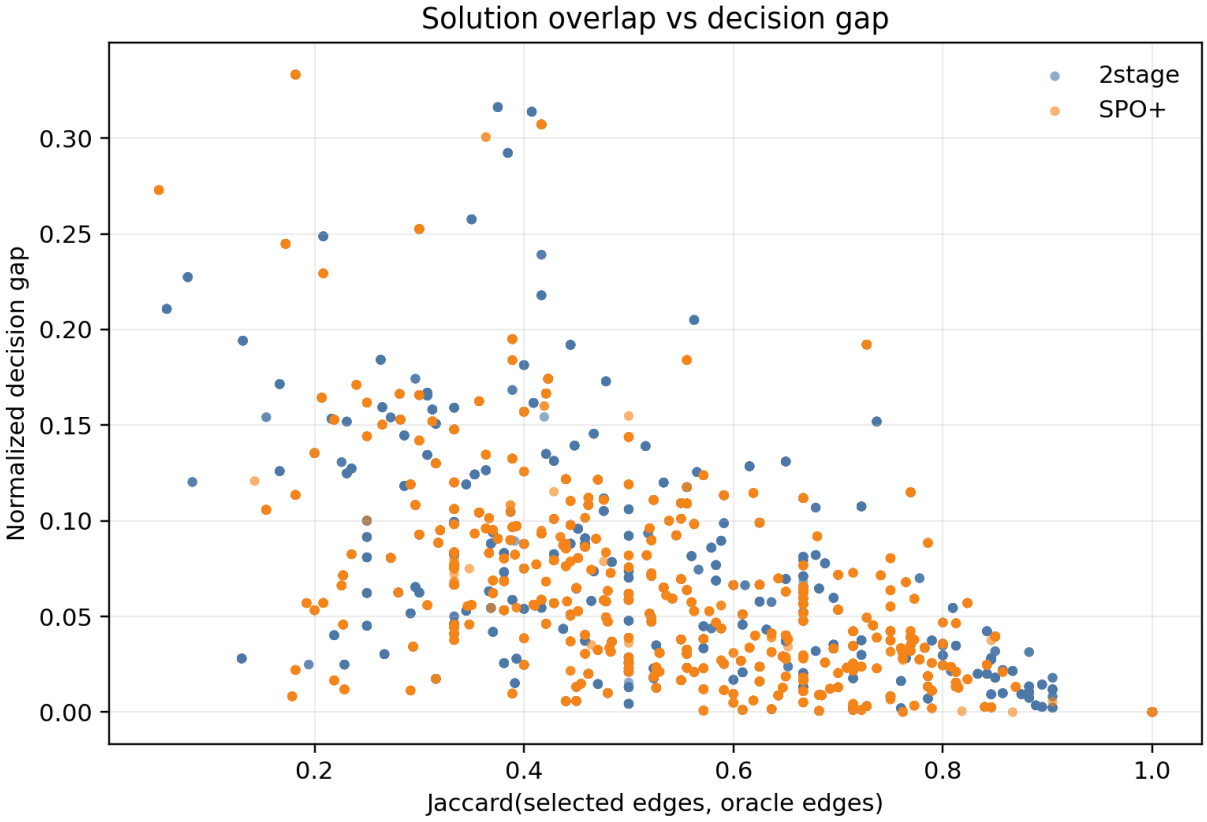

Figure 6: Solution overlap with the oracle versus normalized decision gap.

How to read. The x-axis measures edge-set overlap with the oracle solution, and the y-axis measures normalized gap. Lower-overlap points with small gaps indicate near-optimal alternative exchanges: the selected solution can differ while remaining close in true objective.

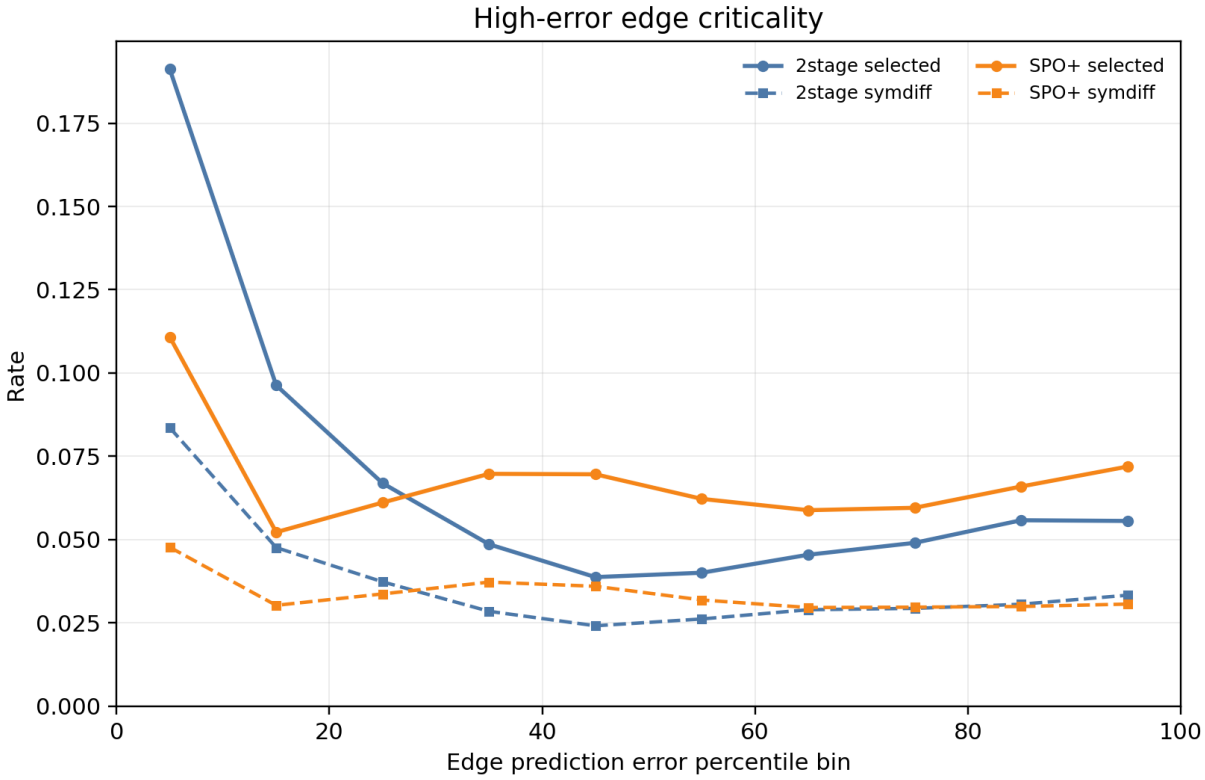

Figure 7: High-error edge selected and symmetric-difference rates.

How to read. The leftmost error-rank bins contain the largest prediction errors. Compare selected-rate and symmetric-difference-rate curves: high-error edges matter most when they enter the selected solution or the predicted-vs-oracle symmetric difference.

Table 4: Decision-level summary on selected Step2b d8 heldout400 cases.

Metric	2stage	SPO+
Mean normalized decision gap	0.06087	**0.04938**
Mean Jaccard with oracle solution	0.60580	**0.63655**
Exact oracle-solution rate	17.81%	**23.58%**
Top-10-error edges selected rate	22.87%	**19.05%**
Top-10-error edges in symmetric difference	9.39%	**7.02%**
Observed 2stage/SPO+ solution relationship		
Same 2stage/SPO+ solution rate	55.08%	
Mean Jaccard between 2stage and SPO+	0.82707	
Median Jaccard between 2stage and SPO+	1.00000	
Any-method near-tie rate, threshold = 0.01	9.17%	
Correlation with normalized gap, Pearson / Spearman		
All-edge MSE	0.095 / 0.023	-0.021 / -0.075
MSE on symmetric-difference edges	0.390 / 0.528	0.272 / 0.372
Top-10-error symmetric-difference rate	**0.570 / 0.667**	**0.557 / 0.639**

4.4 Summary Statistics

4.5 Main Finding

Raw edge-level MSE is weakly correlated with decision gap. In contrast, symmetric-difference-focused measures are much more predictive. This suggests that prediction error matters most when it lands on decision-changing edges, not merely because an edge-level error is large.

4.6 Case Studies

Case A: Bad prediction but irrelevant.

Seed	Graph	2stage gap	All-edge MSE	Top-10-error symdiff rate
41	`G-234.json`	0	19.6328	0.0
41	`G-994.json`	0	18.1776	0.0
25	`G-1516.json`	0	17.7713	0.0

Large prediction errors can occur on edges that do not affect the selected KEP solution. These examples have high all-edge MSE but zero decision gap and zero top-error symmetric-difference involvement.

Figure 8: Large edge-level prediction errors can be non-critical.

How to read. These rows have high all-edge MSE but zero 2stage decision gap. The key column is the top-10-error symmetric-difference rate: it is zero, so the largest prediction errors do not affect the decision-changing edge set.

4.7 Conclusion from Step 3

The first-pass decision analysis supports the advisor's hypothesis. The competitiveness of 2stage does not mean that it predicts every edge accurately. Instead, many large edge-level errors are non-critical because they do not change the selected exchange. In addition, some different KEP

Case B: Different solution but near-optimal.

Seed	Graph	2stage gap	Jaccard with oracle	Same as oracle?
1	G-696.json	0.000644	0.6818	No
1	G-1372.json	0.001174	0.6087	No
1	G-607.json	0.001175	0.7143	No

The selected edge set can differ from the oracle solution while the objective loss remains nearly zero. This indicates that some KEP graphs contain near-optimal alternative exchanges.

Figure 9: Different selected solutions can be near-optimal alternatives.

How to read. These rows are not exact oracle solutions and have only medium Jaccard overlap, but their normalized decision gaps are close to zero. This supports the near-tie story: solution identity can change while true objective remains almost unchanged.

Case C: SPO+ fixes 2stage decision-critical errors.

Seed	Graph	2stage gap	SPO+ gap	2stage top-10 symdiff	SPO+ top-10 symdiff
1	G-392.json	0.313849	0	0.3	0.0
30	G-1560.json	0.292305	0.011964	0.3	0.0
1	G-39.json	0.316226	0.080459	0.2	0.1

SPO+ helps when prediction errors affect decision-changing edges. In these examples, SPO+ substantially reduces decision gap and also reduces high-error symmetric-difference involvement.

Figure 10: SPO+ improves decisions when errors are decision-critical.

How to read. Compare the 2stage and SPO+ gap columns together with the top-10-error symmetric-difference rates. SPO+ reduces the gap precisely in cases where 2stage's large errors enter decision-changing edges.

solutions are near-optimal alternatives under the true objective. SPO+ appears most useful when prediction errors affect edges in the symmetric difference between the predicted and oracle solutions.

5 Overall Conclusions and Next Steps

5.1 Conclusions

This week's work forms a closed loop:

1. **Code validation.** The Step1c SPO+ implementation aligns with PyEPO reference computations on both shortest-path and KEP oracles. The remaining shortest-path full-training SPO+ discrepancy is explained by oracle tie-breaking among multiple shortest paths.
2. **Distributional evidence.** The subset-resampling box plots show how 2stage and SPO+ vary across different 50-graph training subsets on a fixed graph pool.
3. **Mechanistic explanation.** The decision-level analysis suggests why 2stage can remain competitive: many prediction errors are not decision-critical, and some different selected exchanges are near-optimal. SPO+ improves most when errors occur on decision-changing edges.

Table 5: Safe claims supported by the current evidence.

Claim	Evidence	Limitation
Step1c SPO+ implementation is reliable.	PyEPO-reference validation on shortest-path and KEP; KEP Phase 2 trajectory equivalence.	Shortest-path full CSV SPO+ is affected by oracle tie-breaking.
SPO+ improves over 2stage in high-degree subset-resampling settings.	Heldout400 boxplots and paired reductions on fixed graph pools.	Only subset seed varied; not graph seed or label seed.
2stage remains competitive because many errors are non-critical.	Case A/B/C, high-error symdiff rates, and decision-critical MSE correlations.	First-pass evidence on selected Step2b d8 seeds only.

5.2 Limitations

The resampling analysis currently varies only the training subset seed. It does not yet vary graph-generation seed, label-noise seed, or training initialization seed. The decision-level analysis is currently limited to `Step2b d8`, train size 50, heldout400, selected representative seeds, and 2stage vs SPO+. It does not yet include FY, `Step2c d8 eps050`, full graph-seed variation, or adversarial SPO+ solutions y_{adv}.

5.3 Next Steps

The next report should extend the explanation in three directions:

1. Apply the same decision-level pipeline to `Step2c d8 eps050` to check whether the same non-critical-error mechanism holds under multiplicative label noise.
2. Add FY as a third method and compare whether FY-selected solutions are closer to SPO+, 2stage, or the oracle solution.
3. Add y_{adv} analysis to inspect which edges the SPO+ gradient pushes away from or toward.